EDP

EDGE DEVICE MANAGEMENT PLATFORM

Onboard. Update.

Monitor. Operate. Predict.

MVP REQUIREMENTS & ARCHITECTURE

SaaS and Self-Hosted Deployments | Device Onboarding | Monitoring | OTA Updates | User Management

Version 1.5 | 06 November 2026

Edge Device Management Platform (EDP)

Requirements summary, deployment architecture, module scope, security model, diagrams and implementation plan | 06 Nov 2026

1. Product vision

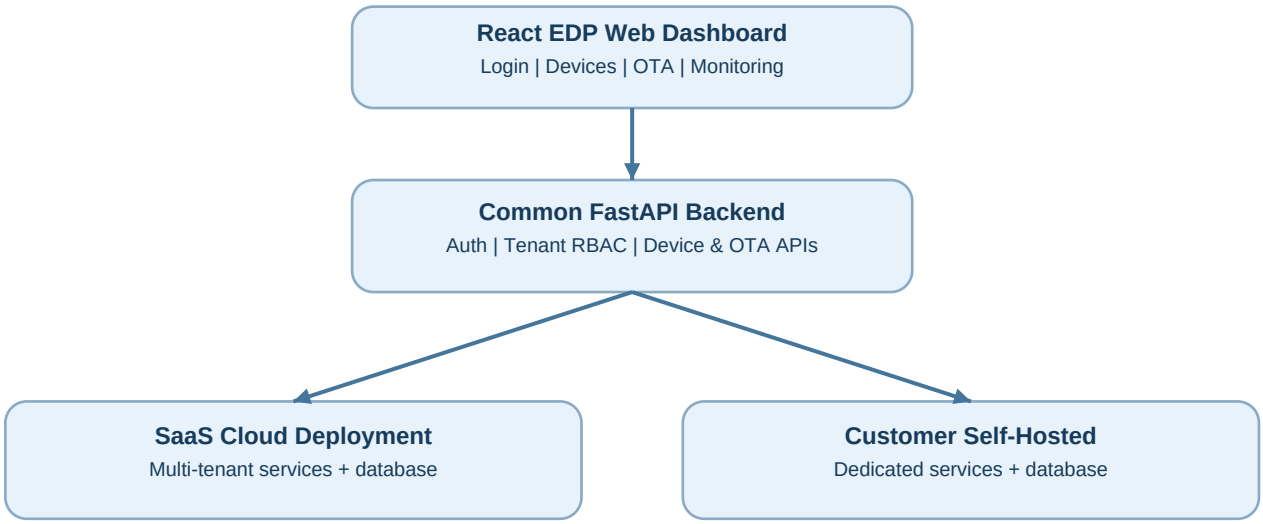
Build a vendor-agnostic platform to onboard, manage, monitor and update Linux/IoT edge devices. Support both cloud-hosted SaaS and customer self-hosted deployments using a shared application codebase and configurable infrastructure.

2. Deployment options

Model	Target use / characteristics
SaaS (managed cloud)	EDP-operated platform; multiple customer organizations with strong logical isolation; centrally managed updates and infrastructure.
Self-hosted (customer environment)	Dedicated deployment on customer servers, private cloud or on-premises; supports isolated-network and offline requirements where specified.

Deployment selection is driven by data residency, security, network connectivity, compliance, operations and scale—not solely by customer size.

3. High-level deployment architecture



The same frontend, API contracts and core business logic serve both deployment modes. Infrastructure, identity-provider settings, network endpoints and operational ownership vary by installation.

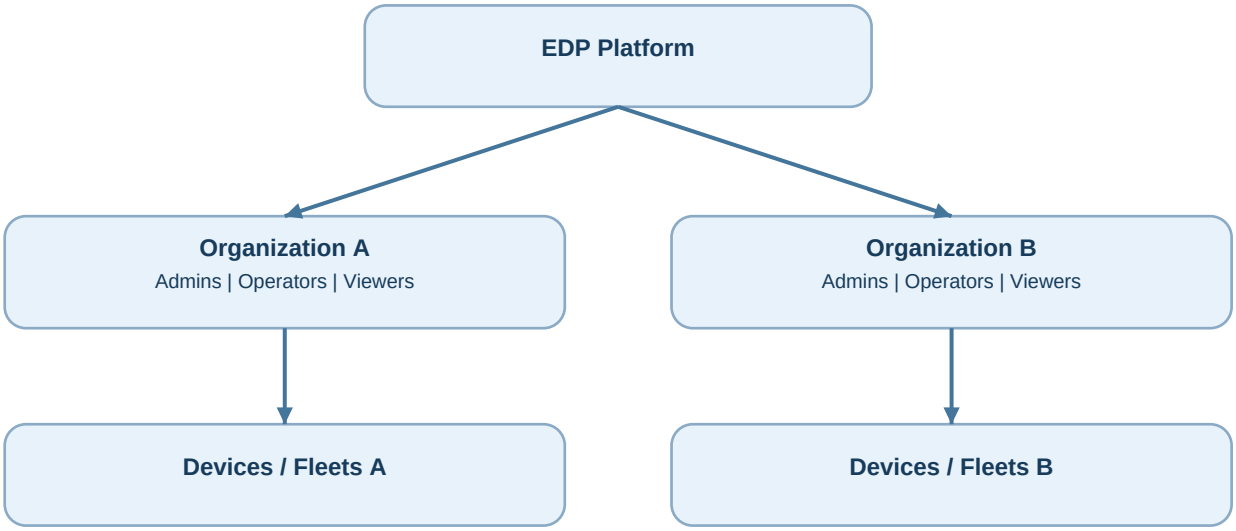
4. Core MVP scope

Priority	Module	MVP capabilities
P0	Authentication & user access	Login, logout, password reset, protected routes, sessions, organization-scoped RBAC.
P0	Dashboard shell	Sidebar, header, routing, overview counters and user profile.
P0	Device inventory	Register/list/view/update/decommission devices; search, filter and status.

Priority	Module	MVP capabilities
P1	Device onboarding	Registration workflow, bootstrap credentials, identity validation, status tracking.
P1	Monitoring	Heartbeat, last seen, CPU/RAM/disk metrics, connectivity status.
P1	OTA management	Update campaign UI, deployment status, result history; rollout engine owned by architect.
P1	Alerts	Basic device alerts, event feed and acknowledgement.

Identity, tenancy and device onboarding

5. Organization and permission hierarchy



SaaS: multiple isolated organizations. Self-hosted: normally a single customer organization, while retaining the same data model. Tenant boundaries must be enforced in API authorization and database queries, not just frontend filtering.

6. Roles and access

Role	Scope / example permissions
Platform Admin	Manage SaaS infrastructure and tenant lifecycle; not automatically entitled to customer data.
Organization Admin	Manage users, roles, fleets and organization settings.
Device Operator	Onboard devices, run approved OTA operations, inspect telemetry.
Viewer	Read-only access to permitted devices and dashboards.

7. Login and identity requirements

• Email/password login, logout, forgot/reset password, session expiry, validation and clear error messages.

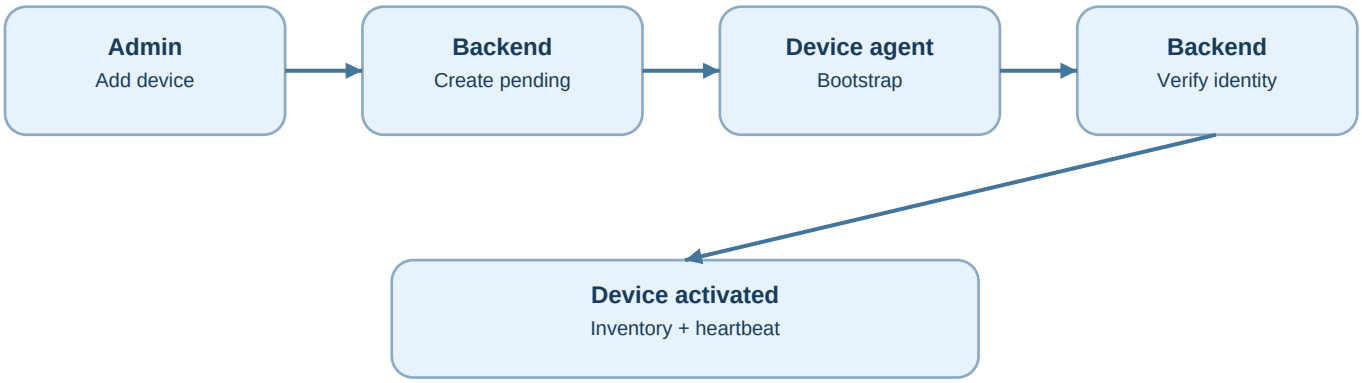
• Secure password hashing, TLS, rate limiting, audit events and protected API endpoints.

• Prefer a mature identity solution or proven auth libraries; do not invent cryptographic/session protocols.

• Enterprise extension: OIDC/SSO and MFA; customer-managed identity providers for self-hosted installations.

• Use least-privilege authorization and organization-scoped access tokens/sessions.

8. Device onboarding workflow



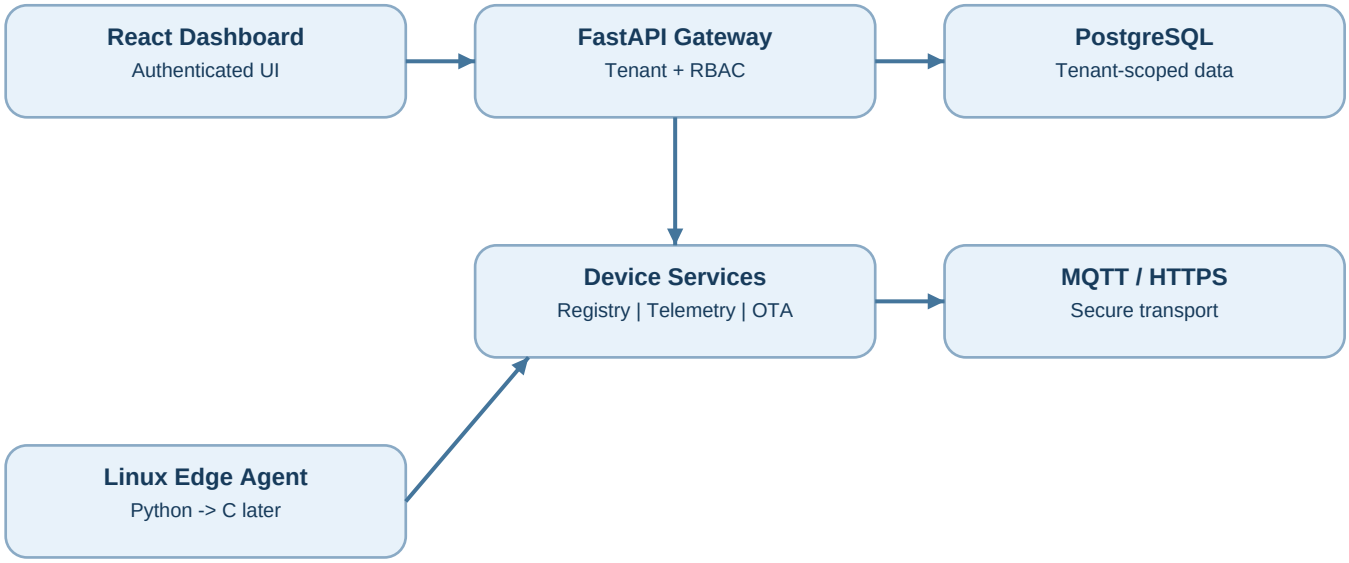
Use short-lived, single-use provisioning credentials; bind registration to an approved device identity; record activation and revocation events. The device agent may begin in Python and later be rewritten in C without changing API contracts.

Implementation reference

9. Suggested technology stack

Layer	Preferred starting point
Web frontend	React + TypeScript, Tailwind CSS, React Router, Recharts
Backend	Python FastAPI, OpenAPI/Swagger, pytest
Persistence	PostgreSQL; organization_id on tenant-owned records
Device transport	HTTPS and/or MQTT; TLS and authenticated device identities
Packaging	Docker images; Compose for simple installations; Kubernetes when justified
CI/CD	GitHub Actions, automated tests, container scanning and release tags

10. Logical platform components



Logical diagram only: gateway, device services and messaging may be separate processes or combined services in the MVP. Agent-to-platform traffic must traverse authenticated, encrypted channels.

11. Initial API contract (draft)

Method / route	Purpose
POST /api/v1/auth/login	Authenticate user (implementation depends on identity solution).
GET /api/v1/me	Current user and authorized organization context.
POST /api/v1/devices	Create pending registration / device record.
GET /api/v1/devices	List authorized organization devices; pagination/filtering.
GET /api/v1/devices/{id}	Device details and last-seen information.
PATCH /api/v1/devices/{id}	Update allowed metadata.

Method / route	Purpose
DELETE /api/v1/devices/{id}	Decommission or soft-delete with audit trail.

All endpoints need input validation, authorization, audit logging for sensitive actions and predictable error formats. Avoid treating CRUD registration as sufficient for trusted device activation.

Delivery plan and decisions

12. Delegation and ownership

Workstream	Fresher (Python / React)	Architect / lead
Login & dashboard	UI, routing, validation, integration	Identity selection, security design, review
Organizations / RBAC	Screens, basic APIs under review	Tenant boundaries, permission policy
Device inventory	CRUD APIs, pages, tests	Schema, onboarding lifecycle
Monitoring / OTA	Status screens, charts, API integration	Agent, MQTT, OTA engine, rollback
Delivery	Unit tests, docs, local setup	Infrastructure, CI/CD, release/security gates

13. Four-week initial implementation sprint

Week	Deliverables	Acceptance checkpoint
1	React setup, login UI, protected routing, dashboard shell	Screens navigable with mocked identity.
2	FastAPI scaffold, PostgreSQL schema, device CRUD	Documented APIs; basic tests and tenant checks.
3	Device registration UI, device detail page, backend integration	End-to-end demo using simulated device.
4	Validation, error states, RBAC checks, testing, review	Working MVP walkthrough and issue backlog.

14. MVP release gates

• No cross-organization data access in API tests; role restrictions enforced server-side.

• Authentication/session handling reviewed; secrets never embedded in frontend code.

• Device registration cannot be activated without verified bootstrap credentials.

• Basic health monitoring and audit logs available; meaningful failure/error states.

• Repeatable local deployment and documented environment configuration.

• Backups, migrations and update/rollback approach documented before production deployment.

15. Decisions to finalize

• Identity provider / authentication library and whether local credentials are required for air-gapped deployments.

• Initial organization model: SaaS multi-tenant and single-organization self-hosted by default.

• Supported onboarding methods for V1: token, claim code, factory provisioned identity, etc.

• Device connectivity: HTTPS, MQTT or both; offline store-and-forward requirements.

• Self-hosted packaging: Docker Compose first; Kubernetes as an enterprise option when required.

• Data retention, tenant isolation strategy, secrets management, backup and upgrade policy.

16. Recommended build sequence

Login + dashboard shell → **organization-aware RBAC** → **device inventory** → **secure onboarding** → **monitoring** → **OTA workflows** → **alerts.** Build locally first with deployment portability and tenancy designed in from day one; add enterprise complexity only when requirements justify it.

Status: architecture and planning reference, not a finalized production security specification.

Device Onboarding: 8 Methods

Four categories | SaaS + self-hosted | proposed product scope

The onboarding framework should support manual, automated and enterprise provisioning without forcing every customer to use a single workflow. All methods converge on the same device identity, organization assignment and lifecycle states.

A. Interactive / user-assisted

1. Manual registration

Admin enters device name, type and serial; device completes activation.

2. QR / claim-code pairing

Operator scans QR or enters short-lived claim code to link a device.

B. Agent / software-assisted

3. Agent install + enrollment token

Install Python/C agent; exchange one-time token for device credentials.

4. CLI / API registration

Register via CLI, REST API or scripted provisioning pipeline.

C. Network / discovery-assisted

5. Local network discovery

Find devices via mDNS/approved network discovery; confirm and claim.

6. Gateway-assisted enrollment

Gateway enrolls downstream devices using scoped delegated authority.

D. Enterprise / bulk provisioning

7. CSV / bulk import

Upload inventory manifest; validate and create pending device records.

8. Zero-touch factory provisioning

Pre-provision identity at manufacturing; auto-claim through trusted bootstrap.

Onboarding Selection & Security

Suggested rollout priorities and required controls

Recommended implementation sequence

MVP / V1

Manual registration; agent token enrollment; CLI/API enrollment

Build and validate end-to-end before adding discovery.

V1.5

QR/claim-code pairing; CSV/bulk import

Add usability and fleet operations.

Enterprise / V2

Local discovery; gateway-assisted; factory zero-touch

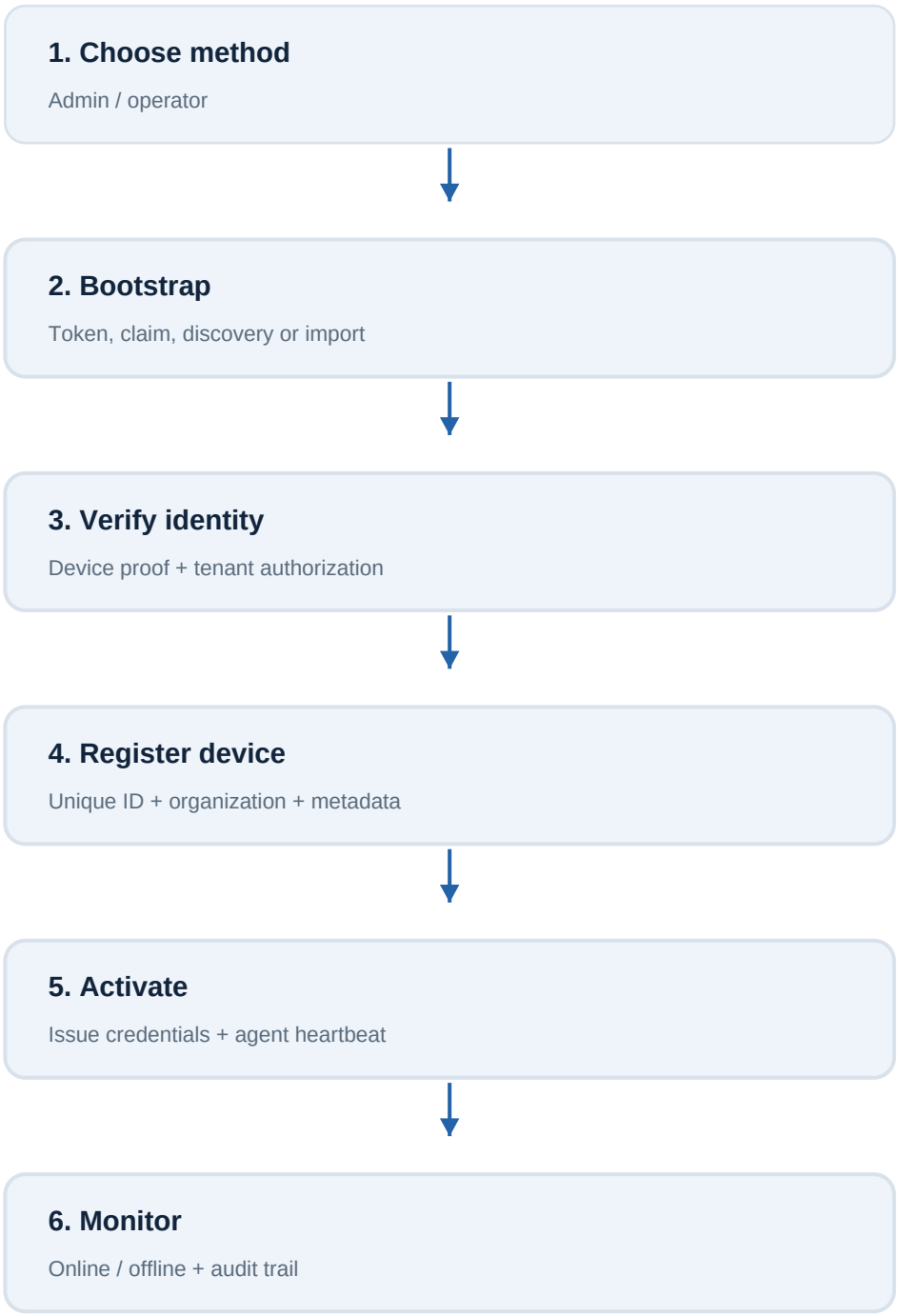
Requires stronger trust, discovery and lifecycle design.

Common security gates for every method

- Authenticate the enrolling user, gateway or manufacturing authority.
- Use short-lived, single-use bootstrap tokens or attested device identities.
- Bind device identity to organization and enforce RBAC on claims.
- Issue rotatable device credentials; never persist plaintext secrets in UI.
- Audit who enrolled what, when, through which method, and the outcome.
- Support revoke, re-enroll, duplicate detection and failed-claim handling.

Unified Enrollment Workflow

One backend state machine across all onboarding methods



Lifecycle states: PENDING -> CLAIMED -> ACTIVATING -> ACTIVE

Failure paths: EXPIRED / REJECTED / QUARANTINED / REVOKED

Reference Screen 1: Onboarding Hub

Concept wireframe - not a final product UI

EDP

- Dashboard
- Devices
- **Onboarding**
- OTA Updates
- Monitoring
- Users
- Settings

Device Onboarding

Select a method to add devices to your organization.

Manual

Enter details

QR / Claim

Pair with code

Agent Token

Install agent

CLI / API

Automate

Discovery

Scan network

Gateway

Enroll children

CSV Import

Upload manifest

Zero-touch

Factory identity

Methods may be enabled by organization policy and deployment mode.

Reference Screen 2: Agent Enrollment

Concept wireframe - enrollment token + agent setup

Enroll a device

Step 2 of 4 | Agent enrollment

1 Method | 2 Details | 3 Bootstrap | 4 Verify

Device name

edge-gateway-001

Device type

Linux / ARM64

Enrollment command (example only)

```
edp-agent enroll --server <EDP_URL>
  --token <ONE_TIME_TOKEN>
```

Generate a short-lived, single-use token after authorization. Display the secret once; never include real credentials in reference screenshots.

Continue

Reference Screen 3: Bulk Import

Concept wireframe - CSV validation + review

Bulk Device Import

Upload CSV > Validate > Review > Import

Drop CSV file here

or select a file from your computer

Preview and validation

Device ID	Type	Organization	Status
gw-001	Linux	Org A	Valid
gw-002	Jetson	Org A	Valid
gw-001	Linux	Org A	Duplicate

2 valid | 1 requires correction | 0 imported

Import valid rows

Reference Screen 4: Device Claim Status

Concept wireframe - enrollment verification and lifecycle

Device Activation

Device: edge-gateway-001

Registration created	Complete
Bootstrap token issued	Complete
Device identity verified	Complete
Credentials provisioned	Pending
First heartbeat received	Waiting

Actions: Retry verification | Revoke enrollment | View audit log

Developer Handoff & API Contract

Implementation guidance for ReactJS + FastAPI

Core backend endpoints (proposed)

GET	/api/v1/onboarding/methods	Enabled methods and policy
POST	/api/v1/devices	Create pending device
POST	/api/v1/enrollments	Start enrollment
POST	/api/v1/enrollments/{id}/claim	Claim / verify proof
GET	/api/v1/enrollments/{id}	Status and audit-safe metadata
POST	/api/v1/enrollments/{id}/revoke	Invalidate pending claim
POST	/api/v1/devices/imports	Validate/import CSV manifest

Fresher ownership

Build onboarding hub, method wizard, agent enrollment screen, bulk import preview, status timeline, form validation, API integration and component tests.

Architect ownership

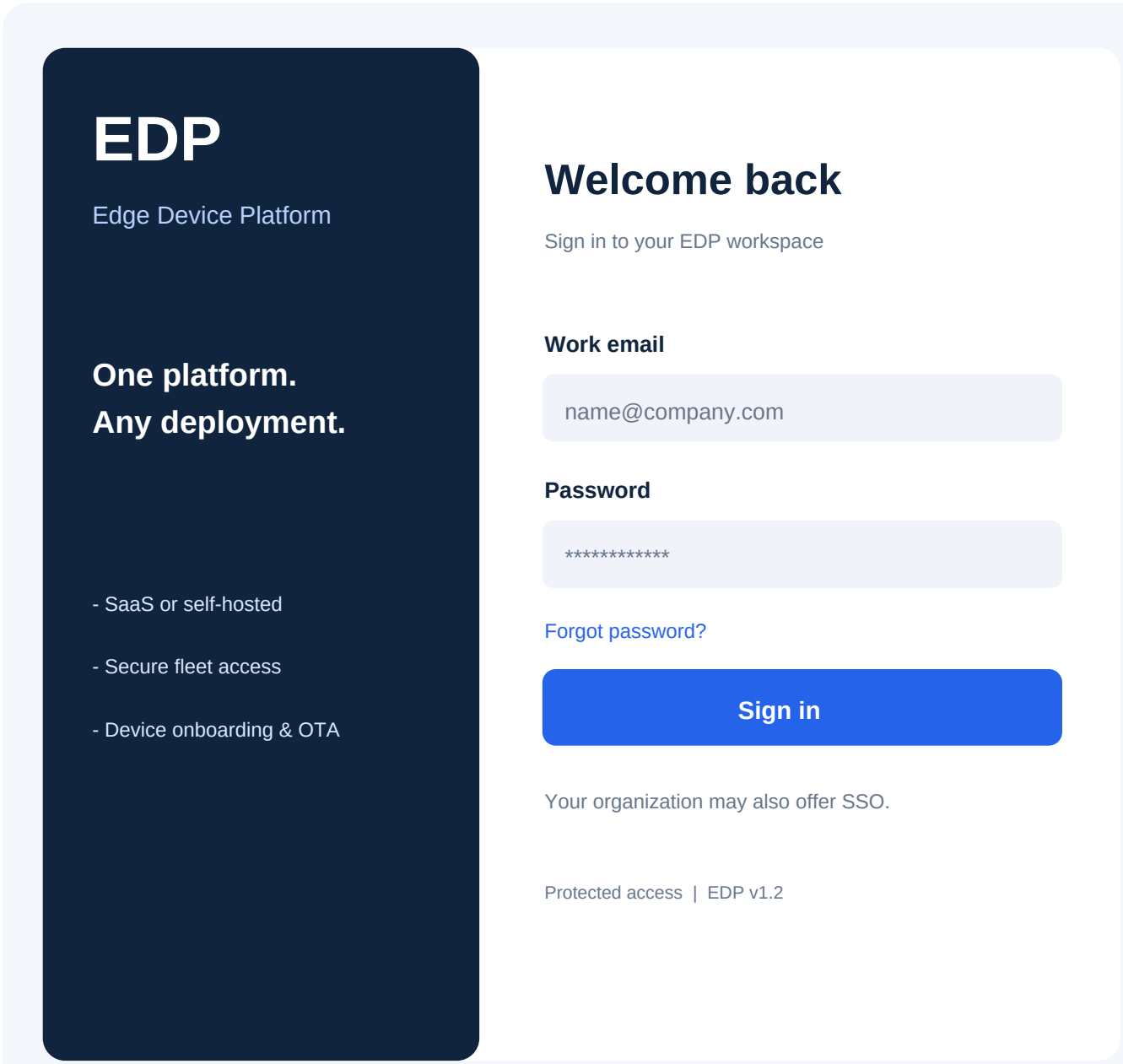
Define trust model, device identity, bootstrap token lifecycle, tenant authorization, secret handling, protocol design, factory provisioning and gateway delegation.

Acceptance criteria

Every enrollment is tenant-scoped, auditable, idempotent where appropriate, duplicate-aware and recoverable after failure. All methods produce a common device identity and status model.

EDP | Login & Authentication

Reference screen and implementation requirements



Screen behavior

- Validate required fields and email format; show generic sign-in errors.
- Redirect authenticated users to the dashboard; protect all private routes.
- Offer password reset and optional enterprise SSO when configured.
- Support accessible labels, keyboard navigation, and responsive layout.

Wireframe for implementation reference; not a finalized brand design.

Login | Functional & Security Specification

Applicable to both SaaS and customer-hosted EDP deployments

Frontend (ReactJS)

- Login, logout, password reset request and reset confirmation screens.
- Form validation, loading states, error states, and accessible controls.
- Protected routing and session-expiry handling; avoid token storage in localStorage.

Backend / Identity

- Use a trusted identity system or well-maintained authentication libraries.
- Prefer secure HttpOnly, Secure, SameSite cookies and CSRF protection as applicable.
- Rate-limit authentication, securely hash passwords, and audit sign-in events.
- Support local accounts initially; enable OIDC/SSO for enterprise deployments.

Organization & RBAC

- Resolve organization membership on the server after authentication.
- Enforce tenant isolation and role permissions on every protected API.
- Roles: platform admin (SaaS), organization admin, operator, viewer.

Acceptance Criteria

- Invalid credentials do not disclose whether an account exists.
- Unauthenticated users cannot access dashboard or device APIs.
- Logout invalidates the session; expiration requires re-authentication.
- Self-hosted mode does not require a connection to EDP SaaS identity services.

Implementation owner: fresher (UI/API integration); architecture and security review: lead.

EDP MVP | End-to-End Gap Review & Screen Reference

Version 1.3 | Supplemental specification to the existing v1.2 requirements. The original 18 pages are retained, followed by this reviewed MVP appendix.

Review verdict

The previous document establishes the architecture and some onboarding/login screens, but it does **not** yet fully specify the complete user journey. Missing screen-level contracts, first-run setup, device detail, OTA workflow, monitoring, alerts, user administration, and operational release gates are addressed here. This is a build-ready design baseline, not proof of implemented functionality.

MVP boundary and definition of done

Area	MVP requirement / acceptance
Deployment	One tested SaaS tenant path and one tested single-customer self-hosted path; same app images; documented environment config.
Identity	Bootstrap admin, login/logout, forgot/reset password, session expiry, RBAC and tenant-scoped APIs.
Devices	Register/enroll one real Raspberry Pi agent, list, view, monitor, decommission and audit.
Connectivity	Agent enrollment via one-time token, periodic heartbeat, TLS, offline/reconnect behavior.
OTA	Upload/register approved artifact, target one device, trigger, track result, recover/rollback on failure.
Operations	Logs, backups/restore rehearsal, secrets, health checks, migrations, and installation instructions.

Onboarding scope: eight planned, three initial

V1: manual registration, agent-token enrollment, and CLI/API registration. Later: QR/claim code, local discovery, gateway enrollment, CSV/bulk import, zero-touch provisioning. All share a device lifecycle and auditable claim API. The later-method screens in the earlier pages are references, not MVP delivery commitments.

Navigation map & critical journeys

User journey	Screens / checkpoints
First installation	Deployment setup -> create initial admin -> login -> dashboard empty state
New device	Onboarding hub -> choose agent token -> create pending device -> copy install command -> activation status -> device detail
Operate device	Device list -> device detail -> metrics / heartbeat / event history -> alerts
Update device	OTA artifacts -> create campaign -> confirm targets -> rollout progress -> result / rollback
Manage access	Users & roles -> invite/create user -> set role -> revoke access -> audit
Recover	Forgot password -> reset -> login; offline device -> reconnect; failed OTA -> retry/rollback

Cross-cutting UX requirements

All data views need loading, empty, error, permission-denied and success states. Tables need search, pagination, filters and tenant-scoped queries. Destructive operations require confirmation. Timestamps use UTC in APIs and local display time in UI. Do not expose secrets after issuance; allow rotation/revocation.

Backend boundaries

React UI -> FastAPI authenticated API -> PostgreSQL for organizations, users, devices, jobs and audit events. MQTT/HTTPS ingestion -> agent heartbeat/telemetry and OTA status. Artifact storage may be object storage or a configured local equivalent. Use background workers for OTA and notification tasks; do not run long jobs in synchronous HTTP requests.

Screen specifications | Access & setup

Reference screen: First-run administrator setup

Route: **/setup (only when uninitialized)**

LEFT NAV / CONTEXT	MAIN SCREEN WIREFRAME
EDP | Dashboard | Devices | Onboarding | OTA | Monitoring | Alerts | Users | Settings	Create organization name | admin email | password | confirm password | deployment mode shown as read-only
Primary actions	Initialize platform; then redirect to login
Required UI states	Setup disabled after initialization; duplicate bootstrap blocked; validation and server failure

Implementation notes: Self-hosted bootstrap must be protected by installation secret or restricted local access. SaaS tenant creation follows controlled signup/invite flow.

Reference screen: Login / forgot password / reset

Route: **/login, /forgot-password, /reset-password**

LEFT NAV / CONTEXT	MAIN SCREEN WIREFRAME
EDP | Dashboard | Devices | Onboarding | OTA | Monitoring | Alerts | Users | Settings	EDP logo | Email | Password | Show password | Sign in | Forgot password | optional SSO entry
Primary actions	Sign in; request reset; set new password; sign out
Required UI states	Invalid credentials; expired link; locked/rate-limited; session expired

Implementation notes: Use secure identity implementation, httpOnly secure cookies where applicable, CSRF protection, server-side authorization. No hardcoded credentials.

Reference screen: Dashboard overview

Route: **/dashboard**

LEFT NAV / CONTEXT	MAIN SCREEN WIREFRAME
EDP | Dashboard | Devices | Onboarding | OTA | Monitoring | Alerts | Users | Settings	Cards: Total / Online / Offline / Pending | Recent devices | Latest alerts | Recent OTA jobs | Onboard device CTA
Primary actions	View device list; start onboarding; open alerts
Required UI states	Empty fleet; no metrics; partial API outage

Implementation notes: Dashboard counts must reflect only authorized organization devices.

Screen specifications | Fleet & enrollment

Reference screen: Device list / inventory

Route: **/devices**

LEFT NAV / CONTEXT	MAIN SCREEN WIREFRAME
EDP | Dashboard | Devices | Onboarding | OTA | Monitoring | Alerts | Users | Settings	Table: Name | Device ID | Model | Org | Status | Last seen | Agent version | Actions; search and filters
Primary actions	Add device; view detail; decommission
Required UI states	Empty, loading, offline, pending, failed, paginated

Implementation notes: Do not mark offline solely from UI clock; backend computes status from heartbeat policy.

Reference screen: Onboarding wizard

Route: **/devices/onboard**

LEFT NAV / CONTEXT	MAIN SCREEN WIREFRAME
EDP | Dashboard | Devices | Onboarding | OTA | Monitoring | Alerts | Users | Settings	Step 1: select method | Step 2: device metadata | Step 3: one-time token/install command | Step 4: awaiting activation | Step 5: success
Primary actions	Generate token; copy command; revoke/retry; finish
Required UI states	Token expired, already used, wrong tenant, device unreachable

Implementation notes: One-time tokens short-lived and hashed at rest; exchange for per-device identity. No secrets in URL or logs.

Reference screen: Device detail

Route: **/devices/:id**

LEFT NAV / CONTEXT	MAIN SCREEN WIREFRAME
EDP | Dashboard | Devices | Onboarding | OTA | Monitoring | Alerts | Users | Settings	Header: name, status, last seen | Tabs: Overview / Metrics / Events / OTA History / Settings | metadata, agent version
Primary actions	Rename; rotate credential; open OTA; decommission
Required UI states	Offline; no telemetry; forbidden; device deleted

Implementation notes: Device identity and organization immutable to unauthorized users; record all lifecycle actions.

Screen specifications | Operations

Reference screen: Monitoring & telemetry

Route: **/monitoring and device Metrics tab**

LEFT NAV / CONTEXT	MAIN SCREEN WIREFRAME
EDP | Dashboard | Devices | Onboarding | OTA | Monitoring | Alerts | Users | Settings	Fleet health summary | CPU, memory, disk time-series | Last heartbeat | date range | per-device selector
Primary actions	Filter device/time; inspect metrics
Required UI states	No data, stale data, disconnected agent

Implementation notes: Specify metric units, retention, aggregation and ingestion limits; timestamp samples at source.

Reference screen: OTA artifact library

Route: **/ota/artifacts**

LEFT NAV / CONTEXT	MAIN SCREEN WIREFRAME
EDP | Dashboard | Devices | Onboarding | OTA | Monitoring | Alerts | Users | Settings	Artifact table: name, version, platform/architecture, checksum, signature/approval, created at
Primary actions	Register/upload artifact; view details
Required UI states	Unsupported architecture; checksum mismatch; upload failure

Implementation notes: Validate compatibility and integrity before rollout. For MVP, one device and one approved artifact is sufficient.

Reference screen: Create OTA campaign & rollout

Route: **/ota/campaigns/new and /ota/campaigns/:id**

LEFT NAV / CONTEXT	MAIN SCREEN WIREFRAME
EDP | Dashboard | Devices | Onboarding | OTA | Monitoring | Alerts | Users | Settings	Choose artifact -> target device -> preflight -> confirmation -> queued/running/succeeded/failed/rolled back timeline
Primary actions	Launch; cancel where safe; retry; rollback
Required UI states	Offline target; insufficient disk; download failure; reboot timeout

Implementation notes: State transitions are server-owned and idempotent; agent reports installation and health-check results.

Screen specifications | Administration

Reference screen: Alerts & events

Route: **/alerts**

LEFT NAV / CONTEXT	MAIN SCREEN WIREFRAME
EDP | Dashboard | Devices | Onboarding | OTA | Monitoring | Alerts | Users | Settings	List: severity | device | alert type | first/last seen | status | assigned/acknowledged | details
Primary actions	Acknowledge; filter; open device
Required UI states	No alerts; duplicate events; insufficient permission

Implementation notes: MVP alert rules: heartbeat missed, low disk, OTA failure. Include deduplication and acknowledgement audit.

Reference screen: Users, roles & invitations

Route: **/settings/users**

LEFT NAV / CONTEXT	MAIN SCREEN WIREFRAME
EDP | Dashboard | Devices | Onboarding | OTA | Monitoring | Alerts | Users | Settings	Users table: email | role | status | last login; invite user form; role picker
Primary actions	Invite; change role; disable user
Required UI states	Invite expired; duplicate email; last admin protection

Implementation notes: Organization Admin cannot grant Platform Admin; all checks server-side.

Reference screen: Organization / platform settings

Route: **/settings**

LEFT NAV / CONTEXT	MAIN SCREEN WIREFRAME
EDP | Dashboard | Devices | Onboarding | OTA | Monitoring | Alerts | Users | Settings	Organization name | heartbeat threshold | retention | deployment version | account profile | change password
Primary actions	Save settings; view deployment info
Required UI states	Validation errors; read-only user; unavailable settings

Implementation notes: For SaaS show tenant-scoped configuration; self-hosted show local deployment settings to authorized admin.

API contract & data model checklist

Domain	Minimum endpoints / records
Identity	POST /auth/login, /auth/logout, /auth/forgot-password, /auth/reset-password; GET /me; users, roles, organizations, sessions
Devices	GET/POST /devices; GET/PATCH/DELETE /devices/{id}; GET /devices/{id}/events; devices, enrollment_tokens, device_credentials
Enrollment	POST /enrollments; POST /agent/enroll; GET /enrollments/{id}; POST /enrollments/{id}/revoke
Telemetry	POST /agent/heartbeat; POST /agent/metrics; GET /devices/{id}/metrics; heartbeat and metric samples
OTA	GET/POST /ota/artifacts; GET/POST /ota/campaigns; GET /ota/campaigns/{id}; POST /agent/ota-status; artifacts, campaigns, targets, transitions
Alerts / audit	GET /alerts; POST /alerts/{id}/ack; GET /audit-events; alert_rules, alerts, audit_events

Core data model / ownership

Organization 1:N Users, Devices, OTA Campaigns and Alerts. Device 1:N Heartbeats, Metrics, Enrollment Attempts and OTA Targets. OTA Campaign 1:N OTA Targets. All tenant-owned rows carry organization_id and use server-verified tenant scope. Keep audit records for privileged actions.

Minimum state machines

Enrollment: pending -> token_issued -> claimed -> active; terminal: expired / revoked / failed. **Connectivity:** online / stale / offline. **OTA:** draft -> queued -> downloading -> installing -> verifying -> succeeded; failures -> failed / rolled_back. Define allowed transitions and retries.

Acceptance tests & release checklist

Test path	Pass condition
First-run	New deployment creates admin once, redirects to login; no unauthenticated setup access afterward.
Tenant isolation	Org A cannot list, query, mutate, or subscribe to Org B devices, metrics, jobs, alerts or users.
RBAC	Viewer cannot onboard, delete, update OTA or manage users; operator/admin permissions verified by API tests.
Enrollment	New Pi enrolls with one-time token; reused/expired token rejected; device receives scoped identity.
Telemetry	Agent heartbeat and CPU/RAM/disk appear in UI; unplug/reconnect changes status correctly.
OTA happy path	Approved matching artifact deploys to one Pi; checksum checked; job reaches succeeded with reported version.
OTA failure path	Bad artifact / interrupted update results in failed state and documented recovery or rollback.
Security	TLS, password reset, brute-force controls, secret rotation, input validation, audit log, dependency scanning.
Deployment	Compose installation tested; SaaS tenant provisioning tested; restart, backup/restore, migration and health check documented.
UX	Every screen includes loading/empty/error/forbidden states, mobile-friendly login and accessible form labels.

Implementation sequencing (fresher vs architect)

Sprint	Fresher ownership	Architect / lead ownership
1	Login, setup, dashboard shell, reusable UI states	Identity choice, bootstrap security, tenancy/RBAC
2	Device list/detail, onboarding wizard, FastAPI CRUD	Enrollment protocol, agent identity, Pi integration
3	Metrics and alert screens, API wiring	Telemetry ingest, status rules, alert processing
4	OTA artifact/job screens, tests, docs	OTA agent/executor, preflight, recovery, security review
5 (hardening)	Bug fixes, E2E test automation, installation UI polish	SaaS + self-hosted validation, backup/restore, release gates

Delivery caveat: Five sprints are planning slices, not a reliable completion estimate for one fresher. A real device OTA loop and security hardening may require more time.

Architecture diagrams | MVP deployment & flow

Common deployment topology

SaaS installation	Self-hosted installation
React frontend + FastAPI + worker + PostgreSQL + MQTT broker + artifact storage (cloud managed or containers)	Same app components and interfaces on customer infrastructure; locally configured storage/broker; no required cloud control plane

Both deployments -> HTTPS/MQTT device ingress -> Python agent on Raspberry Pi -> heartbeat/metrics and OTA status. Agent maintains outbound connectivity where possible.

End-to-end device lifecycle

Step	System action	Visible screen
1. Admin signs in	Identity/session issued, org context resolved	Login -> Dashboard
2. Register device	Pending device and one-time enrollment token	Onboarding wizard
3. Agent enrolls	Token validated, scoped credentials issued	Activation status
4. Device reports	Heartbeat and metrics persisted; device online	Device detail / Monitoring
5. Operator updates	Artifact validated, OTA job sent, result confirmed	OTA campaign / Progress
6. Alert or decommission	Alert generated or device credential revoked	Alerts / Device settings

Decisions still required before coding

Choose identity implementation (OIDC provider or hardened local auth), SaaS tenant provisioning model, device credential format and rotation, MQTT broker, artifact signing/verification, OTA rollback guarantees on Raspberry Pi, metrics retention, and minimum supported self-hosted topology. Treat these as explicit architecture decisions rather than implied implemented features.

EDP MVP v1.4 | User & Role Management

Scope decision: The requirements in this appendix supersede any earlier MVP descriptions of platform administrators, organization administrators, viewers, custom roles, or granular policy configuration. Those features are future enhancements.

One primary administrator is created during first-run setup. The primary administrator manages user accounts and may assign one of three fixed roles: Admin, Manager, Operator. Additional admins are allowed only when explicitly created by the primary administrator. The initial primary administrator cannot be removed or demoted in the MVP.

Fixed role permissions

Action	Admin	Manager	Operator
Manage users / assign roles	Yes	No	No
Configure platform / organization	Yes	No	No
Onboard and edit devices	Yes	Yes	No
Launch / manage OTA updates	Yes	Yes	No
View devices, telemetry, OTA status	Yes	Yes	Yes
Acknowledge alerts	Yes	Yes	Yes
View audit events	Yes	Yes	No

Required screens

1. **First-run admin setup:** only available when the deployment is uninitialized; create the primary administrator securely. 2. **Login:** email/password, validation, errors, logout. 3. **Users:** list, search, role/status badges, add/edit/deactivate. 4. **Add/Edit User:** name, email, role selector, activation status. 5. **My Profile:** profile details and password change.

Security and acceptance rules

The backend enforces every permission on every request; React route/button visibility is only a usability aid. Use hashed passwords, secure sessions, rate limiting, audit logs for user/role changes, and tenant-scoped access. No public self-registration. An admin must not deactivate their own active session or remove the last active administrator. Self-hosted deployments should not require a vendor-controlled admin account.

Reference screen | User Management

EDP / Users	Search users... [+ Add User]
Name / Email	Role Status Actions

Primary Admin / admin@company.test	ADMIN Active Edit
Operations Lead / lead@company.test	MANAGER Active Edit
Device Operator / op@company.test	OPERATOR Active Edit

Add / Edit User modal: Full name [_____] Email [_____] Role [Admin | Manager | Operator] Status [Active | Disabled] [Cancel] [Save]

Future (not MVP): custom roles, fine-grained policy engine, department-scoped permissions, SSO, approval workflows, delegated role administration.